ResearchHorizon

Multi-agent future-research-direction recommendation, calibrated to MRR

The problem. >7M papers/year — researchers struggle to identify *what to study next*. Given a paper, what are the most promising directions it leaves open?

The task MLRC-Bench framing — mask a paper's "*Future Work*" section → recommend a **ranked list** of candidate directions. The held-out text is the ground truth; the system is optimized for **Mean Reciprocal Rank (MRR)**.

> "*Mask the future-work; have **5 LLM agents + a Semantic-Scholar knowledge graph** independently rank candidate directions; fuse them into one list **calibrated to dev MRR**; emit the best-of {agents, KG, fusion} — **never worse than the best single ranker.***"

Complete pipeline

① 5 LLM Agents

gpt5nano · qwen · mixtral · llama17 · llama70 — each emits a ranked `future_directions` list → **rank by MRR, keep top 2**

② Semantic Scholar KG

resolve CorpusId → **pull connected papers** (refs + citations) → entity co-occurrence → Agent-C / white-space S2 source

▼ **cluster the 3 sources (llm1 + llm2 + s2)** ▼

③ Rank Fusion — `consensus.py`

weighted RRF + LLM-agreement(η) + agreement×S2-support(ρ) · **tuned on dev MRR@3** (multi-restart coordinate ascent, warm-started from plain RRF)

▼

④ CV Best-of Fallback — `select_best_strategy_cv`

k-fold CV on dev · emit the **simplest strategy within margin** of the best → **provably ≥ best single source**, robust to the dev/test gap

▼

⑤ Evaluation

MRR@3 vs masked GT · MLRC benchmark vs random baseline · **Direction Cards** (attribution · ablation · logic · faithfulness)

Stages 1–2: candidate sources

① LLM agents `run_llm_selection.py`

5 LLMs each emit ranked `future_directions`; matched to masked GT, scored on the
same 806 papers (fair: misses = 0) → keep **top 2**.

agent	MRR	agent	MRR
llama17	**0.524**	mixtral	0.347
llama70	**0.520**	gpt5nano	0.267
qwen	0.348		

② Semantic Scholar `live S2 API`

Resolve the seed's CorpusId, **pull connected papers** (refs + citations + abstracts),
build an **entity co-occurrence graph**.

> **Agent C** ranks neighbour papers by graph signals (recency/novelty/relevance)
>
> **White-space** generates entities *adjacent to the seed but unexplored* (§7,§9.7)
>
> score = adjacency · 1/(1+support) · gap_boost

Live, SEXIT paper: pulled 23 connected papers incl. "SRH needs of immigrant adolescents" — the paper's documented gap.

Stage 3: rank fusion + safety net

Consensus RRF consensus.py — *"when both LLMs rank a direction near the top it's strong evidence; the S2 graph supports it."*

$$S(c) = \Sigma\, \mathbf{w_m}/(k+r_m(c)) \;+\; \boldsymbol{\eta_1}\cdot A12_top1 + \boldsymbol{\eta_3}\cdot A12_top3 \;+\; \boldsymbol{\rho}\cdot A12_top3\cdot R_{s2}$$

Tuned on dev MRR@3

multi-restart coordinate ascent, **warm-started from plain RRF** → can never lose to RRF on dev. The 3 sources are **clustered** first, so differently-phrased versions of one direction accumulate per-source ranks.

CV best-of fallback the robustness piece

k-fold CV **re-fits each fusion per fold** → a fusion that only wins by *overfitting* dev gets a lower CV score; emit the **simplest** strategy within margin (single < RRF < consensus) → emits the best single LLM, not an overfit fusion.

Explainability — Direction Cards §6

A ranked list isn't enough — a scientist must **trust and verify** each suggestion.
Every output is an auditable **Direction Card** (`explain.py`):

🎯 Attribution
exact per-source RRF decomposition (sums to the score) / linear-SHAP

🔬 Causal ablation
leave-one-signal-out: "*remove S2 → drops to rank 7*"

⚖️ Logical reasoning
defeasible argumentation → *recommend* / *caveat* / *abstain* + trace

📊 Calibration & abstention
calibrated confidence (ECE); "*insufficient evidence*" when thin

✅ Faithfulness check
the shown rationale must match the *actual* ranking drivers

📎 Evidence grounding
links the direction to the `key_gaps` it addresses (quoted spans)

*The **model-risk-aware layer**: borrowed from interpretability (SHAP; Ivanitskiy, Mechanistic Interpretability) — the explanation reflects the **real computation**, not a post-hoc story.*

Why it works — results & rigor

MLRC benchmark (75 test papers, MRR@3)

system	MRR@3	Δ baseline	%ceiling
random baseline	0.07	—	0%
best LLM (llama70/17)	**0.33**	**+340%**	**27%**
consensus RRF	0.32	+339%	27%
S2 (Agent C) alone	0.02	–71%	—
oracle / recall ceiling	1.00		100%

+340% over baseline

candidate **recall ≈ 100%** (oracle=1.0) → the answer is in the pool; the open problem is *ranking*

Metric decides, not the LLM

CV best-of emits the best generaliser — *never < best single source*

Honest finding

fusion ≈ the best single LLM today; the 5 LLMs are *correlated* + S2 is near-random → needs diverse LLM families (GPT+Claude) + a stronger S2

Rigor

temporal split · bootstrap CIs · Bonferroni · k-fold CV · **53 passing tests**

References

- **Cormack et al. 2009** — *Reciprocal Rank Fusion*
- **Sun et al. 2023** — *RankGPT* (listwise)
- **Qin et al. 2023** — *PRP* (pairwise)
- **Cao et al. 2007** — *ListNet* · Burges — *LambdaMART*
- **Cohan et al.** — *SPECTER2 / SciNCL* (embeddings)
- **Lundberg & Lee 2017** — *SHAP* · Ivanitskiy — *Mech. Interp.*
- **Si, Yang & Hashimoto 2024** — *Can LLMs Generate Novel Research Ideas?* (overgenerate→rerank; **S2-API agent**)
- **Baek et al. 2024** — *ResearchAgent* (citation graph + **knowledge graph**)
- **Wang et al. 2023** — *SciMON* (novelty)
- **MLRC-Bench** — baseline-relative scoring
- **Semantic Scholar / OpenAlex** — connected papers

One `rank_fusion/` package — MRR harness · RRF + consensus fusion · learned rankers (pointwise/pairwise/listwise/soft-MRR) · entity + white-space S2 · autonomous candidate agent · Direction-Card explainability · CV best-of selection · MLRC benchmark.